

and predictably improves with increasing scale (Fig. 3, bottom). This confirms our first prediction and supports our alternative explanation that the source of emergent abilities is the researcher’s choice of metric, *not changes in the model family’s outputs*. We also observe that under Token Edit Distance, increasing the length of the target string from 1 to 5 predictably decreases the family’s performance in an approximately quasilinear manner, confirming the first half of our third prediction.

Prediction: Emergent Abilities Disappear With Better Statistics We next tested our second prediction: that even on nonlinear metrics such as accuracy, smaller models do not have zero accuracy, but rather have non-zero above-chance accuracy *commensurate with choosing to use accuracy as the metric*. In order to accurately measure models’ accuracy, we increased the resolution by generating additional test data, and found that on both arithmetic tasks, all models in the InstructGPT/GPT-3 family achieve above-chance accuracy (Fig. 4). This confirms our second prediction. We also observe that as the target string length increases, the accuracy falls approximately geometrically with the length of the target string, confirming the second half of our third prediction. These results additionally demonstrate that the researcher’s choice of metric has the effect that one should predict accuracy to have, i.e., geometric decay with the target length.

4 Meta-Analysis of Claimed Emergent Abilities

Analyzing the GPT family is possible because the models are publicly queryable. However, other model families claimed to exhibit emergent abilities are not publicly queryable, nor are their generated outputs publicly available, meaning we are limited to analyzing the published results themselves [8, 33, 32]. Our alternative explanation makes two predictions.

1. At the “population level” of Task-Metric-Model Family triplets, emergent abilities should appear predominantly on specific *metrics*, not *task-model family* pairs, and specifically with nonlinear and/or discontinuous metrics.
2. On individual Task-Metric-Model Family triplets that display an emergent ability, changing the metric to a linear and/or continuous metric should remove the emergent ability.

To test these predictions, we used to claimed emergent abilities on BIG-Bench [28, 33] due to the benchmark being pertinent and publicly available.

Prediction: Emergent Abilities Should Appear with Metrics, not Task-Model Families If emergent abilities are real, one should expect task-model family pairs to show emergence for all reasonable metrics. However, if our alternative explanation is correct, we should expect emergent abilities to appear only under certain metrics. To test this, we analyzed on which metrics emergent abilities appear. To determine whether a task-metric-model family triplet exhibits a possible emergent ability, we used a metric from previous work [28]. Letting $y_i \in \mathbb{R}$ denote model performance at model scales $x_i \in \mathbb{R}$, sorted such that $x_i < x_{i+1}$, the emergence score is:

$$\text{Emergence Score}\left(\left\{(x_n, y_n)\right\}_{n=1}^N\right) \stackrel{\text{def}}{=} \frac{\text{sign}(\arg \max_i y_i - \arg \min_i y_i)(\max_i y_i - \min_i y_i)}{\sqrt{\text{Median}(\{(y_i - y_{i-1})^2\}_i)}} \quad (1)$$

We found that most metrics used in BIG-Bench have *zero* task-model family pairs that exhibit emergent abilities: of the 39 preferred metrics in BIG-Bench, at most 5 display emergence (Fig. 5A). Many of the 5 are nonlinear and/or discontinuous, e.g., Exact String Match, Multiple Choice Grade, ROUGE-L-Sum (App. A.4). Notably, because BIG-Bench often scores models on tasks using multiple metrics, the *lack* of emergent abilities under other metrics suggests that emergent abilities do not appear when model outputs are scored using other metrics.

Because emergence score only *suggests* emergence, we also analyzed hand-annotated task-metric-model family triplets [32], which revealed emergent abilities appear with 4/39 metrics (Fig. 5B), and 2 metrics account for > 92% of claimed emergent abilities (Fig. 5C): Multiple Choice Grade and Exact String Match. Multiple Choice Grade is discontinuous, and Exact String Match is nonlinear.

Prediction: Changing Metric Removes Emergent Abilities To test our second prediction, we focused on the LaMDA family [30] because its outputs are available through BIG-Bench. For our

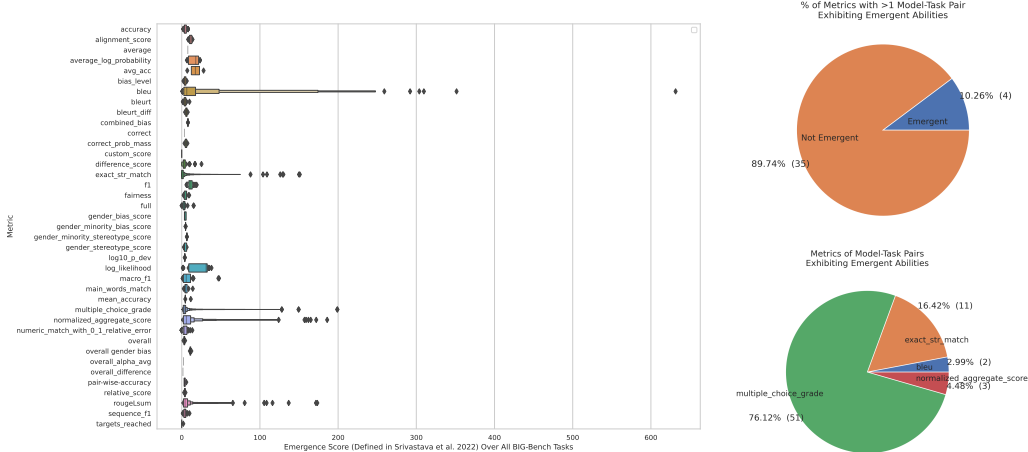


Figure 5: **Emergent abilities appear only for specific metrics, not task-model families.** (A) *Possible* emergent abilities appear with *at most* 5 out of 39 BIG-Bench metrics. (B) Hand-annotated data by [32] reveals emergent abilities appear only under 4 preferred metrics. (C) $> 92\%$ of emergent abilities appear under one of two metrics: Multiple Choice Grade and Exact String Match.

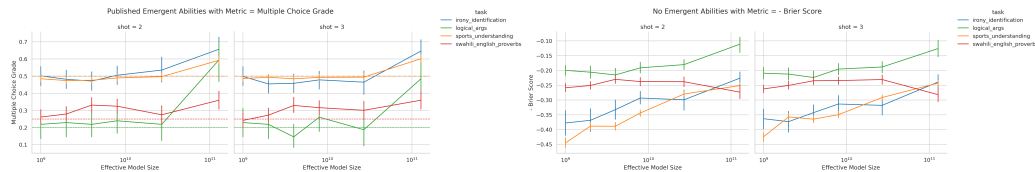


Figure 6: **Changing the metric when evaluating task-model family pairs causes emergent abilities to disappear.** Left: The LaMDA model family displays emergent abilities when measured under the discontinuous Multiple Choice Grade. Right: The LaMDA model family’s emergent abilities disappear when measured under a continuous BIG-Bench metric: Brier Score.

analysis, we identified tasks on which LaMDA displays emergent abilities with Multiple Choice Grade, then asked whether LaMDA still displays emergent abilities on the same tasks with a different BIG-Bench metric: Brier Score [2]. Brier Score is a strictly proper scoring rule for predictions of mutually exclusive outcomes; for a binary outcome, the Brier Score simplifies to the mean squared error between the outcome and its predicted probability mass. LaMDA’s emergent abilities on the discontinuous Multiple Choice Grade disappeared when we changed the metric to the continuous Brier Score (Fig. 6). These results support our alternative explanation that emergent abilities are induced by the chosen metric.

5 Inducing Emergent Abilities in Networks on Vision Tasks

To demonstrate how emergent abilities can be induced by the researcher’s choice of metric, we show how to produce emergent abilities in deep networks of various architectures: fully connected, convolutional, self-attentional. We focus on vision tasks because abrupt transitions in vision models’ capabilities have not been observed to the best of our knowledge; this is one reason why emergence in large language models is considered so interesting. For the convolutional example, see App. B.

Emergent Reconstruction of CIFAR100 Natural Images by Nonlinear Autoencoders We first induce an emergent ability to reconstruct images in shallow (i.e., single hidden layer) nonlinear autoencoders trained on CIFAR100 natural images [19]. To emphasize that the sharpness of the metric is responsible for emergent abilities, and to show that sharpness extends to metrics beyond Accuracy, we intentionally define a discontinuous metric that measures a network’s ability to reconstruct

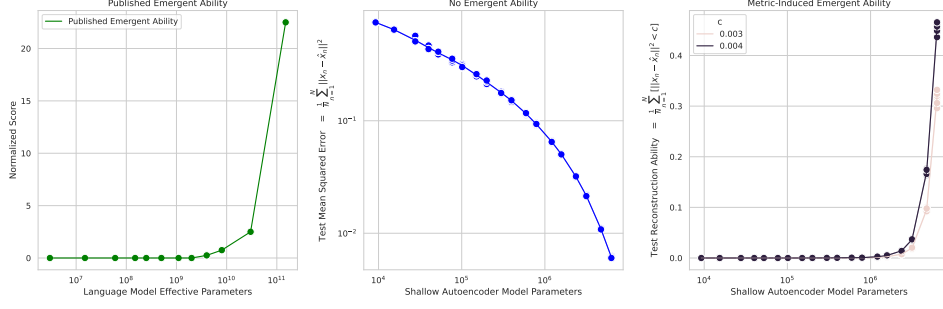


Figure 7: **Induced emergent reconstruction ability in shallow nonlinear autoencoders.** (A) A published emergent ability at the BIG-Bench Periodic Elements task [28]. (B) Shallow nonlinear autoencoders trained on CIFAR100 [19] display smoothly decreasing mean squared reconstruction error. (C) Using a newly defined Reconstruction_c metric (Eqn. 2) induces an unpredictable change.

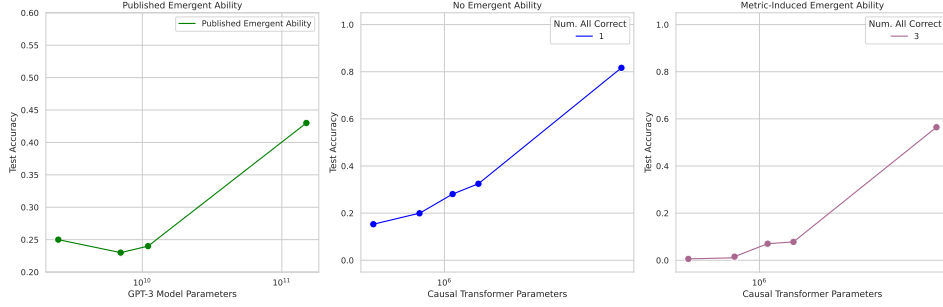


Figure 8: **Induced emergent classification ability in autoregressive Transformers.** (A) A published emergent ability on the MMLU benchmark [8]. (B) Autoregressive transformers trained to classify Omniglot images display increasing accuracy with increasing scale. (C) When accuracy is redefined as classifying *all* images correctly, a seemingly emergent ability appears.

a dataset as the average number of test data with squared reconstruction error below threshold c :

$$\text{Reconstruction}_c(\{x_n\}_{n=1}^N) \stackrel{\text{def}}{=} \frac{1}{N} \sum_n \mathbb{I}[\|x_n - \hat{x}_n\|^2 < c] \quad (2)$$

where $\mathbb{I}(\cdot)$ denotes an indicator variable and $\hat{x}_n$ is the autoencoder’s reconstruction of x_n . The autoencoder family displays smoothly decreasing squared reconstruction error as the number of bottleneck units increases (Fig. 7B). Under our newly defined Reconstruction_c metric and for particular choices of c , the autoencoder family exhibits a sharp and seemingly unpredictable image reconstruction ability (Fig. 7C) that qualitatively matches published emergent abilities (Fig. 7A).

Emergent Classification of Omniglot Characters by Autoregressive Transformers We next induce emergent abilities in Transformers [31] trained to autoregressively classify Omniglot handwritten characters [20], in a setup inspired by recent work [5]: Omniglot images are embedded by convolutional layers, then sequences of embedded image-image class label pairs are fed into decoder-only transformers. We measure image classification performance on sequences of length $L \in [1, 5]$, again via *subset accuracy*: 1 if all L images are classified correctly (Fig. 8B), 0 otherwise. Causal transformers display a seemingly emergent ability to correctly classify Omniglot handwritten characters (Fig. 8C) that qualitatively matches published emergent abilities (Fig. 8A).

6 Related Work

Srivastava et al. [28] observed that while accuracy at a particular task can empirically appear sharp and unpredictable, cross entropy does not; the authors then hypothesized that emergent abilities may be partially attributed to the metric. Our paper converts their discussion into precise predictions,

then quantitatively tests the predictions to reveal that: metric choice is likely wholly responsible for emergent abilities; well-known and widely-used metrics (including ones already used by [28]) capture graded improvements; emergent abilities do not appear only for tasks involving multiple steps, and indeed appear most commonly on the discontinuous Multiple Choice Grade; metric choice can be used to induce emergent abilities in a novel domain (vision) in diverse architectures and tasks.

Caballero et al. [4] explain emergence by assuming a piece-wise power law functional form; under this view, emergent abilities are real, caused by a change in the governing power law. In contrast, our work suggests that emergent abilities are induced by the researcher, even under a single power law. Michaud et al. [25] posit that emergent abilities may be real under strong data assumptions.

7 Discussion

Our paper presents an alternative explanation for claimed emergent abilities of large language models. For a fixed task and a fixed model family, the researcher can choose a metric to create an emergent ability or choose a metric to ablate an emergent ability. Ergo, *emergent abilities may be creations of the researcher’s choices, not a fundamental property of the model family on the specific task*. We emphasize that nothing in this paper should be interpreted as claiming that large language models *cannot* display emergent abilities; rather, our message is that previously claimed emergent abilities in [3, 8, 28, 33] might likely be a mirage induced by researcher analyses.

Our paper has several implications. Firstly, a task and a metric are distinct and meaningful choices when constructing a benchmark. Secondly, when choosing metric(s), one should consider the metric’s effect on the per-token error rate and adapt their measuring process accordingly, e.g., if one chooses accuracy, one should make sure to have sufficient data to accurately measure accuracy to avoid the risk of drawing invalid scientific conclusions. Thirdly, when making claims about capabilities of large models, including proper controls is critical. In this particular setting, emergent abilities claims are possibly infected by a failure to control for multiple comparisons. In BIG-Bench alone, there are ≥ 220 tasks, ~ 40 metrics per task, ~ 10 model families, for a total of $\sim 10^6$ task-metric-model family triplets, meaning probability that *no* task-metric-model family triplet exhibits an emergent ability by random chance might be small. Fourthly, scientific progress can be hampered when models and their outputs are not made public for independent scientific investigation.

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